

1 SUPPLEMENTARY TABLES AND FIGURES

1.1 Tables

Table S1. Performance metrics of the Random Forest model across all groups.

Group	AUC	Accuracy	Recall	Precision
Students	0.996 $\pm$ 0.004	0.962 $\pm$ 0.018	0.925 $\pm$ 0.040	0.969 $\pm$ 0.028
ARS	0.996 $\pm$ 0.007	0.947 $\pm$ 0.047	0.958 $\pm$ 0.052	0.944 $\pm$ 0.072
TASS	0.986 $\pm$ 0.008	0.924 $\pm$ 0.043	0.855 $\pm$ 0.093	0.903 $\pm$ 0.104

0.8

Note: AUC = Area Under the ROC Curve; Accuracy = proportion of correctly classified cases; Recall = sensitivity (true positive rate); Precision = positive predictive value. Values are reported as mean $\pm$ SD across 5-fold stratified cross-validation (Random Forest classifier, 50 decision trees, random_state = 42). This model was trained and evaluated on the same five variables used to derive the well-being clusters; the resulting near-perfect values reflect this circularity by construction and do not constitute external validation of the cluster solution. See Section 3.1 of the main manuscript for full interpretation.

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